CHAPTER 4 · DERIVATIONS

WORK, ENERGY AND POWER

4
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 GRAVITATIONAL POTENTIAL ENERGY NEAR THE EARTH

4.2(A)

RESULT $\Delta U = mgh$

1. Lift a mass m slowly (constant velocity, no change in KE) through a height h near the Earth's surface, where g is constant.
2. The applied force balances the weight: $F = mg$.
3. Work done $W = Fh = mgh$.
4. This work is stored as gravitational potential energy: **$\Delta U = mgh$**.

2 KINETIC ENERGY

4.2(B)

RESULT $K = \frac{1}{2}mv^2$

1. A constant resultant force F accelerates a mass m from rest through a displacement s.
2. Work done $W = Fs = (ma)s$.
3. From $v^2 = u^2 + 2as$ with $u = 0$: $as = \frac{v^2}{2}$.
4. $W = m\frac{v^2}{2}$. The work done becomes kinetic energy: **$K = \frac{1}{2}mv^2$**.

3 WORK-ENERGY THEOREM

4.2(D)

RESULT $\Delta K = F_R s$

1. A resultant force F_R changes the speed of a mass m from u to v over a displacement s.
2. $v^2 = u^2 + 2as$, so $as = \frac{v^2 - u^2}{2}$.
3. $F_R s = mas = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$.
4. **Work done by the resultant force = change in kinetic energy**, $\Delta K = F_R s$.

4 POWER

4.3(A)

RESULT $P = Fv$

1. Power is the rate of doing work: $P = \frac{dW}{dt}$.
2. For a force F along the displacement, $dW = F\,ds$.
3. $P = F\frac{ds}{dt} =$ **Fv** (if F is at angle θ to v, $P = Fv\cos\theta$).